

BE Ist Year Test III (CS A& B)
IET-DAVV
Basic Electronics

Note: Attempt all questions carrying five marks each.

Q1) Write short notes on the following. (5×2)

- i) Varactor diode ii) Tunnel Diode

Q2) Compare between CB, CE and CC configuration of an NPN transistor.

Q3) Given an NPN Transistor for which $\alpha = 0.98$. A common emitter connection is used and $V_{CC} = 12V$, $R_C = 4.0 K\Omega$. What is the minimum base current required in order that the transistor enter its saturation region?